

Student Grades Are Hard to Predict: A Machine Learning Analysis on Synthetic Educational Data

Barna Berekméri
Eötvös Loránd University

Abstract—This study investigates student performance prediction using a synthetic educational dataset containing over 1,000 records and multiple academic and behavioral features. A Random Forest regression model [1] is then trained to estimate students' final grades based on variables such as attendance and previous academic performance. Although the dataset's synthetic nature results in very weak correlations between input features and the target variable, the analysis illustrates the challenges of predictive modeling in noisy environments and highlights the importance of feature relevance in machine learning workflows. The study also demonstrates how visualization and feature importance techniques support understanding model behavior and dataset limitations.

Introduction

Predicting student performance has become an increasingly important topic in modern educational research, as accurate forecasts can support early intervention, personalized instruction, and resource allocation. The rise of machine learning has enabled new analytical approaches for modeling academic outcomes by learning patterns from historical data. Regression models, combined with feature analysis are widely used tools for understanding how various

behavioral and academic factors contribute to students' final results.

The aim of this study is to construct a complete data-processing and predictive modeling pipeline using a publicly available synthetic student performance dataset *Student Performance Predictions* from Kaggle [2]. The workflow includes data cleaning, outlier correction, missing value imputation, categorical feature encoding, and the preparation of training and test sets. A Random Forest regression model is then trained to estimate students' final grades, followed by a detailed evaluation of model accuracy through metrics and visualizations such as distribution plots,

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predicted-versus-actual comparisons, and feature importance analysis.

The dataset contains over 1000 generated student records with variables such as attendance rates, study hours, extracurricular involvement, and parental support. It intentionally includes data errors, missing values, and noise to simulate realistic data-processing challenges. However, exploratory analysis reveals very low correlations between most input features and the FinalGrade target variable. This lack of meaningful relationships poses a significant challenge for predictive modeling.

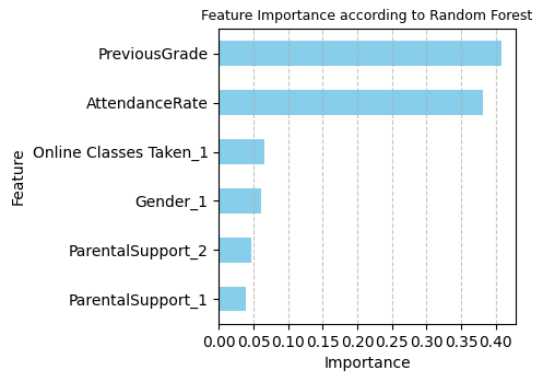


FIGURE 1. Feature importance scores from the Random Forest model. Only PreviousGrade and AttendanceRate show notable importance, while all other features contribute minimally.

Methodology

Machine Learning Model

To predict students' final grades, a Random Forest Regressor was selected as the primary machine learning model. Random forests are ensemble methods that construct multiple decision trees and aggregate their outputs, offering robustness against noise and overfitting. They are particularly well suited for heterogeneous tabular datasets, where relationships between variables may be nonlinear or weakly structured.

The Random Forest model was configured using the following hyperparameters:

- **n_estimators = 300**: number of trees in the ensemble, providing stable predictions.
- **max_depth = None**: trees grow fully unless constrained by other stopping conditions.
- **min_samples_split = 2** and **min_samples_leaf = 1**: default values that allow trees to model fine-grained patterns.

- **max_features = "sqrt"**: a widely used setting that increases tree diversity and reduces correlation between trees.
- **bootstrap = True**: enables sampling with replacement for each tree, improving generalization.
- **n_jobs = -1**: uses all available processor cores to speed up training.

Evaluation Metrics

Model performance was assessed using three standard regression metrics that capture different aspects of prediction quality.

Mean Absolute Error (MAE)

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_{\text{true},i} - y_{\text{pred},i}| \quad (1)$$

MAE measures the average magnitude of prediction errors in the original units (grade points). Lower values indicate better performance.

Mean Squared Error (MSE) and Root Mean Squared Error (RMSE)

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_{\text{true},i} - y_{\text{pred},i})^2 \quad (2)$$

$$RMSE = \sqrt{MSE} \quad (3)$$

These metrics penalize larger errors more strongly and reflect overall prediction stability.

Coefficient of Determination (R^2)

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_{\text{true},i} - y_{\text{pred},i})^2}{\sum_{i=1}^n (y_{\text{true},i} - \bar{y})^2} \quad (4)$$

The R^2 score indicates how much variance in the target variable is explained by the model. Values close to 1 represent strong predictive power, whereas values near or below 0 indicate poor performance.

Results and Discussion

The Random Forest regression model was evaluated on the test set after completing all preprocessing steps. The results indicate very limited predictive performance. The Mean Absolute Error (MAE) is approximately **7.7 grade points**, and the coefficient of determination (R^2) is **-0.11**, meaning the model performs worse than a simple baseline that always predicts the average final grade. This suggests that the model is unable to learn meaningful relationships between the features and the FinalGrade target.

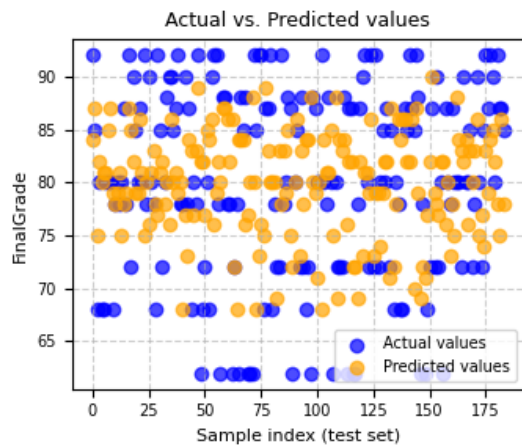


FIGURE 2. Actual vs. predicted FinalGrade values on the test set. The Random Forest model produces tightly clustered predictions, indicating weak correspondence with the true grade distribution.

Visualization further confirms this. The predicted values form a narrow distribution centered around the dataset mean (≈ 80), while the actual grades span a broader range. Predicted-versus-actual plots show minimal alignment, indicating systematic underfitting. Feature importance scores are uniformly low, with no variable contributing substantially to the prediction.

These findings align with the correlation analysis, which shows near-zero correlations between all input features and the FinalGrade. The synthetic dataset—containing noise, missing values, and intentionally weak structure—does not provide the statistical signals needed for effective regression modeling. This highlights a fundamental limitation: even well-designed machine learning pipelines cannot compensate for datasets that lack meaningful predictive information.

Conclusion

This study implemented a complete data-processing and machine learning workflow to predict students' final grades using a synthetic educational dataset. Despite thorough preprocessing and the use of a Random Forest regression model, the predictive performance remained weak. The model's high error and negative R^2 score indicate that the available features do not contain sufficient information to explain or predict the FinalGrade variable.

The core limitation lies in the dataset itself. The intentionally noisy and weakly correlated synthetic

features do not establish meaningful relationships with the target, making accurate regression prediction statistically infeasible. Nevertheless, the project successfully demonstrates the full pipeline of data cleaning, modeling, evaluation, and visualization.

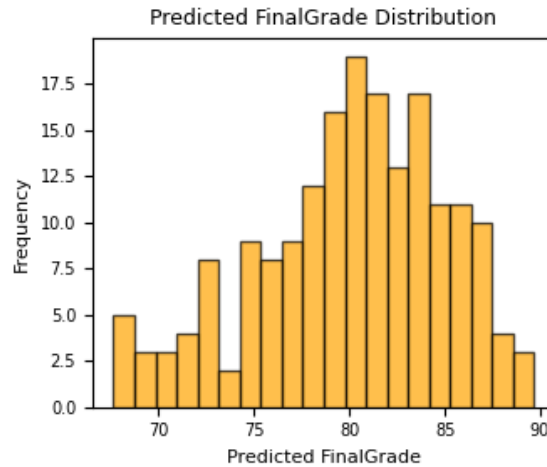


FIGURE 3. Distribution of predicted Final Grade values (y_{pred}).

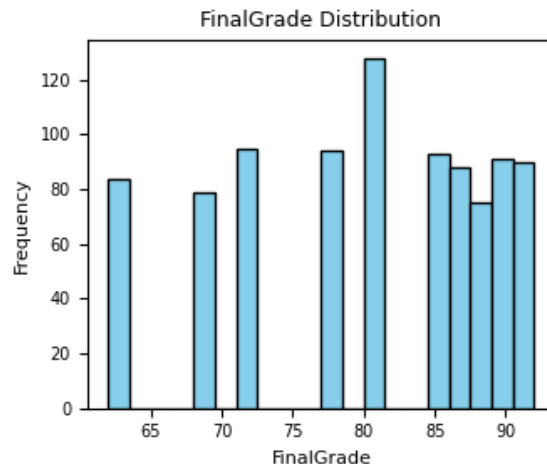


FIGURE 4. Distribution of actual Final Grade values.

REFERENCES

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